



SPONTOM ENTERPRISE PRIVATE LIMITED

Build with AI (Google) • Track 4

Agri AI Network

From scattered farm data
to one field-level decision.

An interoperable AI advisory layer for India's smallholder farmers,
with district intelligence for agriculture officers.

GEMINI-GROUNDED

OFFICER-REVIEWED

STATE-AGNOSTIC

A concept proposal and working prototype by Spontom Enterprise Private Limited

DPIIT Recognised • Visakhapatnam • September 2026

2. Farm intelligence

Groundnut • 2.4 acres • Anantapur

Demo mode • sample data

Stage: pod development • 80 days after sowing • rainfed • conventional

Farm Health

73

/ 100

Moderate

Vegetation (30%)

51

NDVI 0.32 vs ~0.62 expected at flowering pegging

Soil (25%)

70

Low organic carbon (0.38%) • Low available nitrogen (176 kg/ha)

Weather (20%)

100

No significant weather risk in the 7-day forecast

Water (15%)

96

59 mm rain in 14 days vs ~61 mm crop need; rainfed

Crop condition (10%)

55

Crop Doctor: moderate severity (Early Leaf Spot (Cercospora arachidicola); AI-assisted, unverified)

Method farm_health_rules_v1: weighted sum of observed factors; missing factors are re-weighted, never imputed.

Weather

Temperature

26.6 °C

Humidity

72%

Rain, last 14 days

59 mm

Rain forecast, 3 days

5 mm



Open-Meteo forecast API
(ECMWF/GFS/ICON blend)

• observed 5 min ago

Live

Soil

pH

6.4

Organic carbon

0.38%

N / P / K (kg/ha)

176 / 14 / 212

Clay / sand / silt

30 / 47 / 23%

NPK, pH, OC:

AP e-Crop booking + Soil Health Card
export (sample format)

• observed 3 days ago

Sample • synthetic

Soil Health Card style record mapped by
the state adapter

Texture:

ISRIC SoilGrids 2.0 (modelled, 250 m, 0-15
cm)

• observed 7 years ago

Live

Satellite

Crop greenness
(NDVI)

0.32

Δ 30 days

+0.02



Google Earth Engine - Sentinel-2 SR

• observed 17 days ago

Live

Live prototype: Farm Health for a sample 2.4-acre groundnut field in Anantapur, AP

The data exists. It rarely meets at the field decision.

Weather, soil, satellite and state records sit in separate systems; the farmer gets generic advice.

One small field, many disconnected signals.

~86%

holders
small/marginal

2.4

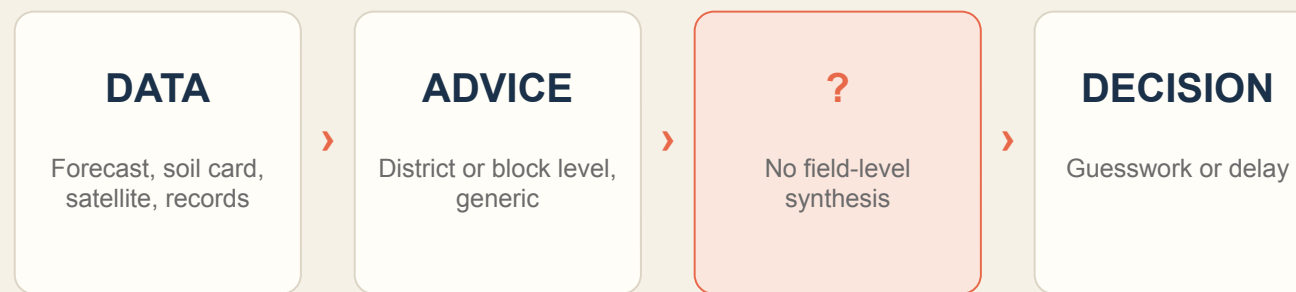
acres (demo field)

0.32

NDVI vs ~0.62
expected

Lakshmi's groundnut is half as green as expected for its stage. Her soil card, forecast and satellite view never meet in one answer.

THE LAST-MILE GAP



FOCUSED PROBLEM

Farmers lack one timely advisor that combines their own field's data and explains itself.

Consequence Late disease response, mistimed irrigation and soil decline; officers see risk only after it spreads.

[1] Agriculture Census 2015-16, via PIB (2023). Field figures: sample farmer record; NDVI is live Sentinel-2 over the sample field polygon (14 Sep 2026).

One advisor for the farmer's own field, grounded in data.

Signals become a transparent score; Gemini explains and recommends; a human releases the advice.

01

FARM INTELLIGENCE

Weather, soil card and Sentinel-2 NDVI become an explainable Farm Health score.

02

GROUNDNUT ADVICE

Gemini reasons only over that context: observed, interpreted, action. English, Hindi, Telugu.

03

HUMAN REVIEW

An Agriculture Officer approves each AI output; Crop Doctor names the nearest agri office.

TODAY'S ADVISORY

GEMINI • SCHEMA-CHECKED

Groundnut • pod development • day 80 • rainfed

1

Observed: NDVI 0.32 vs ~0.62 expected (Sentinel-2, 14 Sep).

2

Observed: organic carbon 0.38%, N 176 kg/ha (soil card, sample).

3

Within 48 h: scout the field for Early Leaf Spot symptoms.

4

Within 48 h: ask the KVK or officer to confirm on site.

5

Regenerative: return groundnut residue to the soil after harvest.

Risk: high • confidence 55% • awaiting review

LISTEN (TTS)

Excerpt of a real advisory from gemini-2.5-flash (prompt advisory_v1), 30 Sep 2026. The soil record is a labelled synthetic sample.

From field signal to reviewed action in seven steps.

Rules where rules suffice, Gemini where reasoning helps, a person where consequences matter.



TIER 0 • RULES

Farm Health and crop scores are transparent rules. Missing factors are re-weighted, never imputed.

TIER 1 • BOUNDED AI

Gemini advisory, crop reasons and Crop Doctor; each record stores model and prompt version.

TIER 2 • HUMAN

Officer approves or rejects. Low-confidence or severe Crop Doctor results go to expert review.

DESIGN RULE The farmer sees AI output labelled as a draft awaiting review; nothing is auto-approved.

Farm Health weights: vegetation 30% • soil 25% • weather 20% • water 15% • crop condition 10% (PRD §11; services/api/app/intelligence/health.py).

Design for Lakshmi first; connect the officer above her.

Exactly two app roles (PRD §7): Farmer and Agriculture Officer. KVKs and states connect through data.

5. Crop Doctor

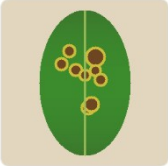
Gemini - gemini-2.5-flash

AI-assisted preliminary assessment from a photo - not a certified diagnosis.

Upload crop photo

Use sample leaf photo

Analyse photo



Potential condition

Early Leaf Spot (Cercospora arachidicola)

Confidence: 80% Severity: moderate leaf

Visible symptoms

- circular brown spots on the leaf
- yellow halos surrounding the brown spots
- spots of varying sizes
- scattered distribution of spots

Other possibilities: Late Leaf Spot (Phaeoisariopsis personata) (70%)

Recommended next steps

- Scout other plants in the field for similar symptoms to assess the spread
- Monitor the progression of the spots on affected leaves
- Consult with a local agriculture officer or KVK for a confirmed diagnosis and specific management recommendations

Nearest agriculture support

Google Maps - live

Sri Krishnadevaraya College Of Agricultural Sciences · Agriculture office · 20.3 km by road, ~36 min drive

Agriculture Office · Agriculture office · 30 km by road, ~53 min drive

Deputy director Of Agriculture · Agriculture office · 33 km by road, ~63 min drive

Google Maps Places (Text Search) + Routes API

AI draft - awaiting Agriculture Officer review

AI-assisted preliminary assessment, not a certified diagnosis. Confirm with your local agriculture officer or KVK before acting, especially before any chemical treatment.

DX-8fe4f56968 - model gemini-2.5-flash - prompt diagnosis_v1 - The image displays characteristic circular brown spots with yellow halos on a groundnut leaf, which are typical symptoms of fungal leaf spot diseases. These conditions are common during the pod development stage in groundnut.

LAKSHMI DEVI • PRIMARY USER (SAMPLE)

2.4 acres • rainfed groundnut • Anantapur, Andhra Pradesh

01

FARMER

Primary user

Needs one clear next action in Telugu, by voice, with sources shown.

02

AGRI OFFICER

District view

Needs district and block risk, hotspots and a review queue, not raw feeds.

03

KVK

Expert referral

Confirms Crop Doctor cases; Maps shows the nearest office and drive time.

04

STATE DEPT.

Data owner

Keeps its own records; joins through a declarative adapter, not a rebuild.

CONSTRAINT → DESIGN RESPONSE

LITERACY

voice • 3 languages

DATA GAPS

source + age badges

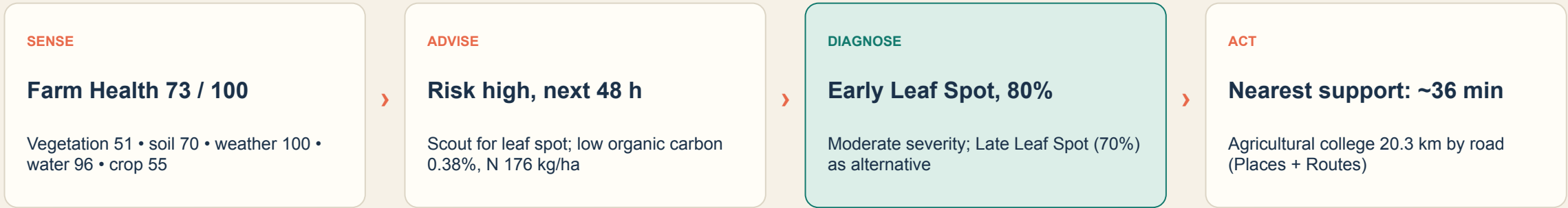
AI ERROR

officer + KVK check

Personas are sample records from the AP/MH/PB adapters. Screenshot: Crop Doctor on the synthetic sample leaf, live Gemini + Google Maps.

Lakshmi's field, end to end, in one continuous demo.

PRD §43 Scenario A: Andhra Pradesh • Anantapur • groundnut • 2.4 acres. Sample farmer, live data feeds.



OFFICER VIEW

Anantapur is the AP hotspot.

Synthetic BigQuery indicators:

+

18 disease / stress alerts

✓

Farm Health 56, water stress 0.70

↗

Lakshmi's advisory in review queue

NEXT SEASON

Bajra 96 • foxtail millet 96 • jowar 91 • green gram 86 • red gram 85 (rule score, explained by Gemini).

Values from the running prototype on 30 Sep 2026; live weather and NDVI change daily. District indicators are labelled synthetic samples.

Not another portal: a field layer that joins them.

India's public systems each solve a valuable part; Agri AI Network composes them per field.

APPROACH	WHAT IT CONTRIBUTES	REMAINING GAP FOR THE FIELD DECISION
Kisan Call Centre	Toll-free expert answers in 22 languages [4]	Answers a call; does not see the field's own data
GKMS agromet	District / block weather-based advisories, IMD with ICAR [3]	Block level, not field level; no photo or soil card link
Soil Health Card	Lab-tested N, P, K, pH, organic carbon per sample [5]	Static card, not joined with weather or crop stage
Digital Agri Mission	AgriStack, Krishi DSS and soil maps as national DPI [2]	Infrastructure layer; farmer-facing advice is left to apps
Agri AI Network	Field context + Gemini advice + photo check + officer review	Prototype on sample state data; not yet field-validated

DISTINCTIVE CAPABILITY

Consumes public-system data through state adapters instead of replacing them.

[2] PIB, Digital Agriculture Mission (2024). [3] PIB, GKMS (2023). [4] PIB, Kisan Call Centre (2021). [5] Soil Health Card portal.

Bounded Gemini on a canonical, state-agnostic model.

Every data block carries source and timestamp; AI returns schema-validated JSON or falls back to rules.

01	CHANNELS	Farmer web app (en / hi / te, voice) • Officer dashboard • Next.js 16
02	STATE ADAPTERS	AP e-Crop, MahaDBT, Punjab land-record formats → one canonical schema
03	DATA LAYER	Open-Meteo • Soil Health Card + SoilGrids • Sentinel-2 via Earth Engine • BigQuery
04	AI LAYER	Gemini 2.5 Flash on Vertex AI • Pydantic-validated JSON • model + prompt version
05	DELIVERY	FastAPI + both apps on Cloud Run • private Cloud Storage • Secret Manager

GOOGLE AI INTEGRATION MAP

- Gemini 2.5 Flash: advisory, Q&A, Crop Doctor (multimodal)
- Speech-to-Text, Text-to-Speech, Translation
- Earth Engine • Maps Places / Routes • BigQuery

PENDING

Firestore + Firebase Auth: **SQLite fallback today**

REAL VS DEMO GATE

Probed live	Each integration is tested with a real call before it is badged Live
Labelled sample	State exports, district indicators, sample leaf: synthetic in data + UI

[6] Sentinel-2 SR Harmonized on Earth Engine. [7] Open-Meteo; ISRIC SoilGrids. Status from /api/system/status, local run, 30 Sep 2026.

Advice that shows its evidence and waits for a human.

Gemini never invents measurements; the officer, not the model, releases consequential advice.

AI DRAFT

AWAITING AGRICULTURE OFFICER REVIEW

Evidence first Observed data, AI interpretation and action are separate fields, each citing its source.

Contestable The officer approves or rejects with a note; the farmer sees the status.

DO NOT DO

- **Certified diagnosis or pesticide prescriptions**
- **Invent or impute missing measurements**
- **Collect identity documents or bank details**

SAFETY BY DESIGN

GROUNDING Structured field context only; validated JSON, else a rules fallback.

UNCERTAINTY Confidence, missing information and data-age notes in every advisory.

CROP DOCTOR “Potential condition” wording; low confidence or high severity → expert.

PROVENANCE Source, timestamp and a Live / Sample badge on every data block.

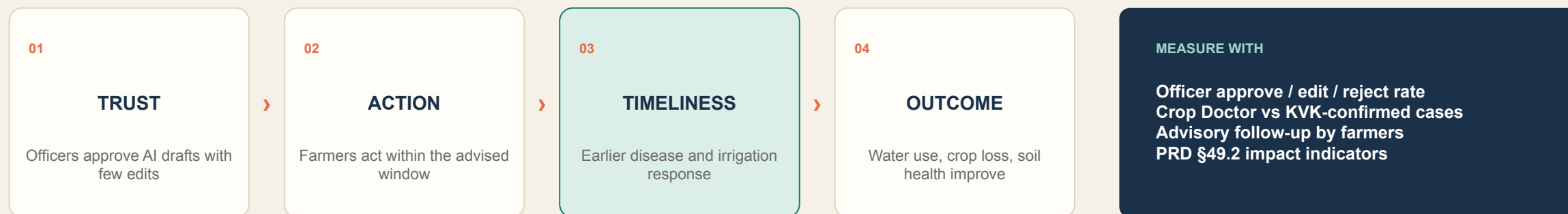
TRACEABILITY Each AI record stores model name, model version and prompt version.

SECURITY Keys in Secret Manager; crop photos in a private bucket; no secrets in git.

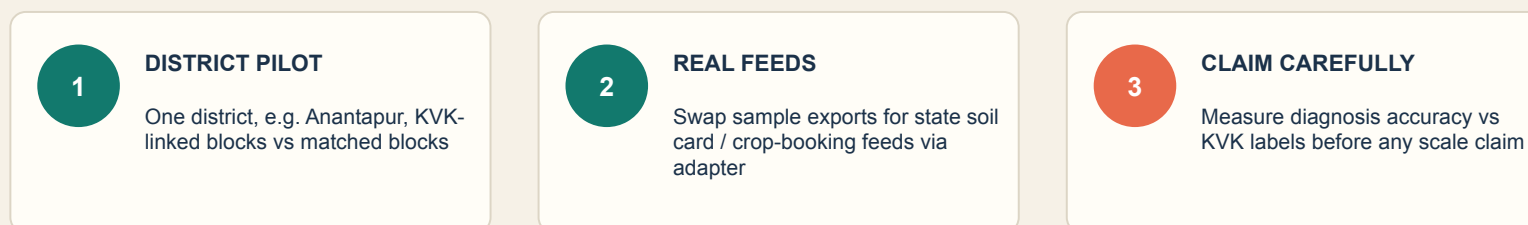
Known gaps: no authentication yet (Firebase Auth planned); Hindi / Telugu strings need native-speaker review (README, Known gaps).

Earn the right to scale: one district, then one state.

The first proof is not app usage; it is timelier, correct action on the field.



VALIDATION GATES



BUDGET ROUTE

Candidate route: Digital Agriculture Mission (₹2,817 cr, Krishi DSS component) with state co-funding; to confirm with the department. [2]

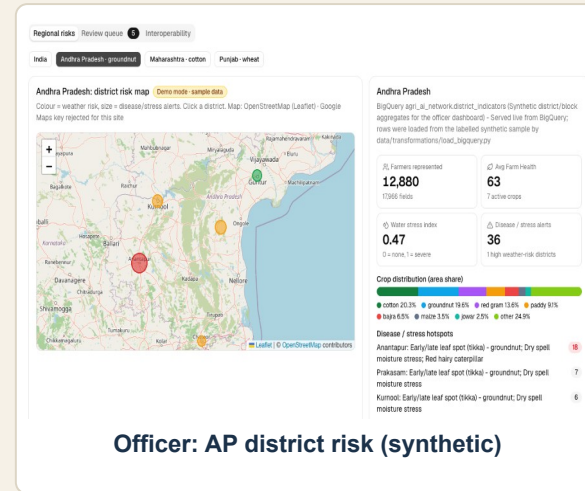
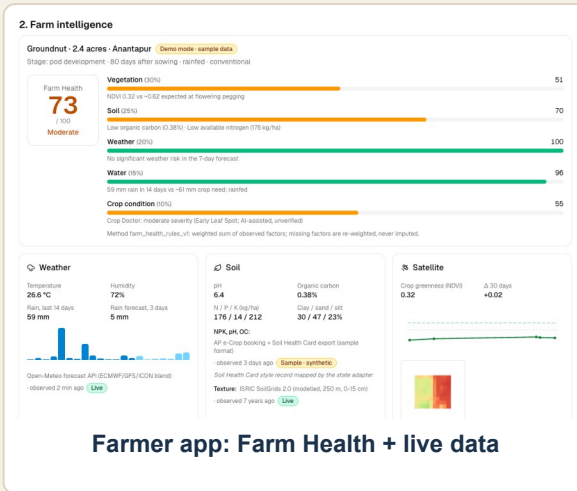
OWNERSHIP + SCALE

Owner	State Department of Agriculture
Channel	Agriculture Officers + KVKs + agromet units
Gate	Data-sharing MoU • language QA • accuracy

Pilot design and measures are proposals to validate with a state partner; no field results exist yet. Scale path follows PRD §50 (district → state → national).

Working today, end to end, on Google Cloud.

Real screenshots from the running prototype, 30 Sep 2026: sample farmer, live Gemini, weather and NDVI.



EVALUATION ALIGNMENT

25%

AI / TECHNICAL EXECUTION

Gemini text + multimodal, schema-validated and versioned; Earth Engine, Speech, Translation

20%

PROBLEM-SOLUTION FIT

PRD \$8 journey end to end: field data → advisory → photo check → action

20%

DEPTH & REACH ACROSS INDIA

3 states via declarative adapters; English, Hindi, Telugu; voice in and out

20%

DEPLOYABILITY & SCALABILITY

Three Cloud Run services, one deploy script, API-first canonical schema

15%

IMPACT POTENTIAL

~86% of holders are small or marginal [1]; view scales district → state

TRY IT

Farmer app

<https://agri-ai-network-farmer-web-847963771142.asia-south1.run.app>

Officer dashboard

<https://agri-ai-network-officer-dashboard-847963771142.asia-south1.run.app>

API (+ /docs)

<https://agri-ai-network-api-847963771142.asia-south1.run.app>

Source

<https://github.com/ipavanreddy/agri-ai-network>

Evidence and public-system sources

Numbered references correspond to citations on the core slides.

- [1] PIB: Livelihood of Farmers (2023)** Ministry of Agriculture & FW, 14 Mar 2023: per Agriculture Census 2015-16, small and marginal holders are about 86% of operational holders.

[OPEN SOURCE ↗](#)

- [2] PIB: Digital Agriculture Mission (2024)** Cabinet approval, 2 Sep 2024: ₹2,817 crore outlay; DPIs AgriStack, Krishi Decision Support System and Soil Profile Mapping.

[OPEN SOURCE ↗](#)

- [3] PIB: Gramin Krishi Mausam Sewa (2023)** IMD with ICAR renders district / block-level agrometeorological advisory services to farmers.

[OPEN SOURCE ↗](#)

- [4] PIB: Kisan Call Centre (2021)** “Addressing Queries of Farmers in 22 languages”, 3 Aug 2021: replies in 22 official languages.

[OPEN SOURCE ↗](#)

Data, platform and project sources

Datasets and services the prototype uses; full list in THIRD_PARTY.md.

[5]	Soil Health Card portal	Government of India scheme; N, P, K rating thresholds used in the Farm Health soil factor. Sample records follow its format.	OPEN SOURCE ↗
[6]	Sentinel-2 SR Harmonized	Copernicus Sentinel-2 surface reflectance on Google Earth Engine; field-mean NDVI / NDMI time series.	OPEN SOURCE ↗
[7]	Open-Meteo; ISRIC SoilGrids 2.0	Live forecast, 14-day history and soil moisture (CC BY 4.0); modelled soil texture at 250 m (soilgrids.org, CC BY 4.0).	OPEN SOURCE ↗
[8]	Agri AI Network PRD + repository	docs/PRD.md (scenarios, Farm Health weights, evaluation weights), README and the source code behind every prototype number.	Source repo ↗